Hands-on U5 keyed RDP



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Introduction

Play with OEM keys !

- Purpose of this hands-on is to understand how to program and use the OEMxKEYS to do regression from any RDP level (including level 2)
- We will program 2 different OEM1KEY and OEM2KEY with specific values
- Then we will use these keys to regress the chip

Hands-on

NUCLEO-U575ZI-Q Prerequisite

Based on : STM32Cube_FW_U5_V1.x.x\Projects\NUCLEO-U575ZI-Q\Examples\GPIO\GPIO_IOToggle_TrustZone

- Configure option byte according to this example :

  TZEN=1
  SECWM1_PSTRT=0x0
  SECWM1_PEND=0x7F
  SECWM2_PSTRT=0x7F
  SECWM2_PEND=0x0

- Flash GPIO_IOToggle_TrustZone binaries provided.

Software Prerequisite

- STM32Cubeprogrammer version 2.9.0 (or upper)
- STLink firmware V3J9M3

Prerequisite

Setup of the board

- Set the option byte :
  - STM32_Programmer_CLI.exe -c port=SWD mode=UR -ob displ
  - STM32_Programmer_CLI.exe -c port=SWD mode=UR -ob TZEN=1
  - STM32_Programmer_CLI.exe -c port=SWD mode=UR -ob SECWM1_PSTRT=0x0 SECWM1_PEND=0x7F SECWM2_PSTRT=0x7F SECWM2_PEND=0x0
- Flash the binaries
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug –d \Training\MOOC_U5\MOOC1\Handson\GPIO_IOToggle_TrustZone_NonSecure.elf
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug –d \Training\MOOC_U5\MOOC1\Handson\GPIO_IOToggle_TrustZone_Secure.elf
- Update STlink firmware if needed :

..\Drivers\FirmwareUpgrade\STLinkUpgrade.jar

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File Edit Format View Help

```
Option byte settings :
STM32_Programmer_CLI.exe -c port=SWD mode=UR  -ob displ
STM32_Programmer_CLI.exe -c port=SWD mode=UR  -ob TZEN=1
STM32_Programmer_CLI.exe -c port=SWD mode=UR  -ob  SECWM1_PSTRT=0x0 SECWM1_PEND=0x7F SECWM2_PSTRT=0x7F SECWM2_PEND=0x0

Flash the binaries
STM32_Programmer_CLI.exe -c port=SWD mode=UR -d \Training\MOOC_U5\MOOC1\Handson\GPIO_IOToggle_TrustZone_NonSecure.elf
STM32_Programmer_CLI.exe -c port=SWD mode=UR -d \Training\MOOC_U5\MOOC1\Handson\GPIO_IOToggle_TrustZone_Secure.elf

Update STlink firmware
```

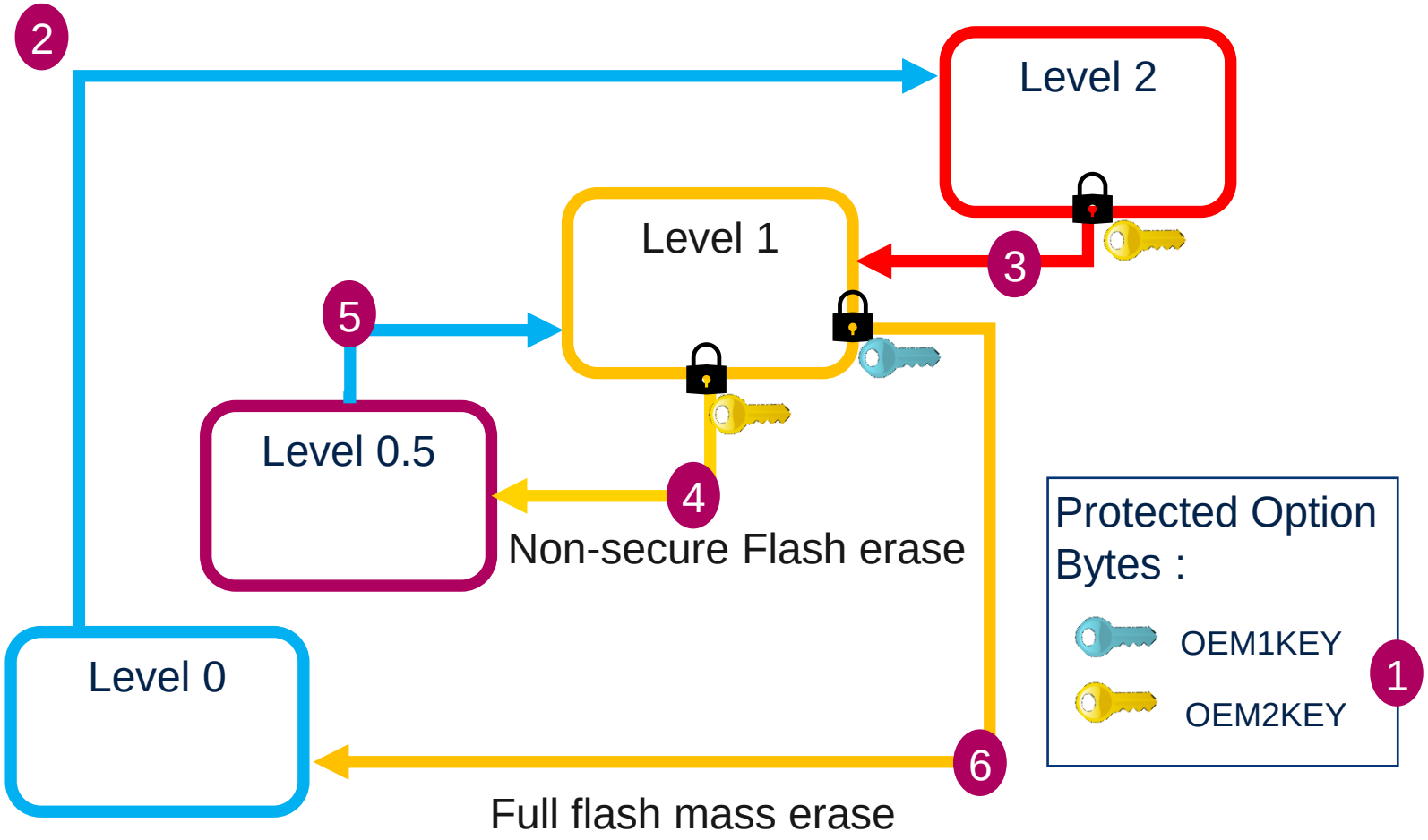
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Command Prompt



U5 Keyed RDP

RDP transitions scenario



Step 1 : Provisioning

Step 2 : RDP level 2

Step 3 : RDP level 1

Step 4 : RDP level 0.5
reflash non-secure App

Step 5 : RDP level 1

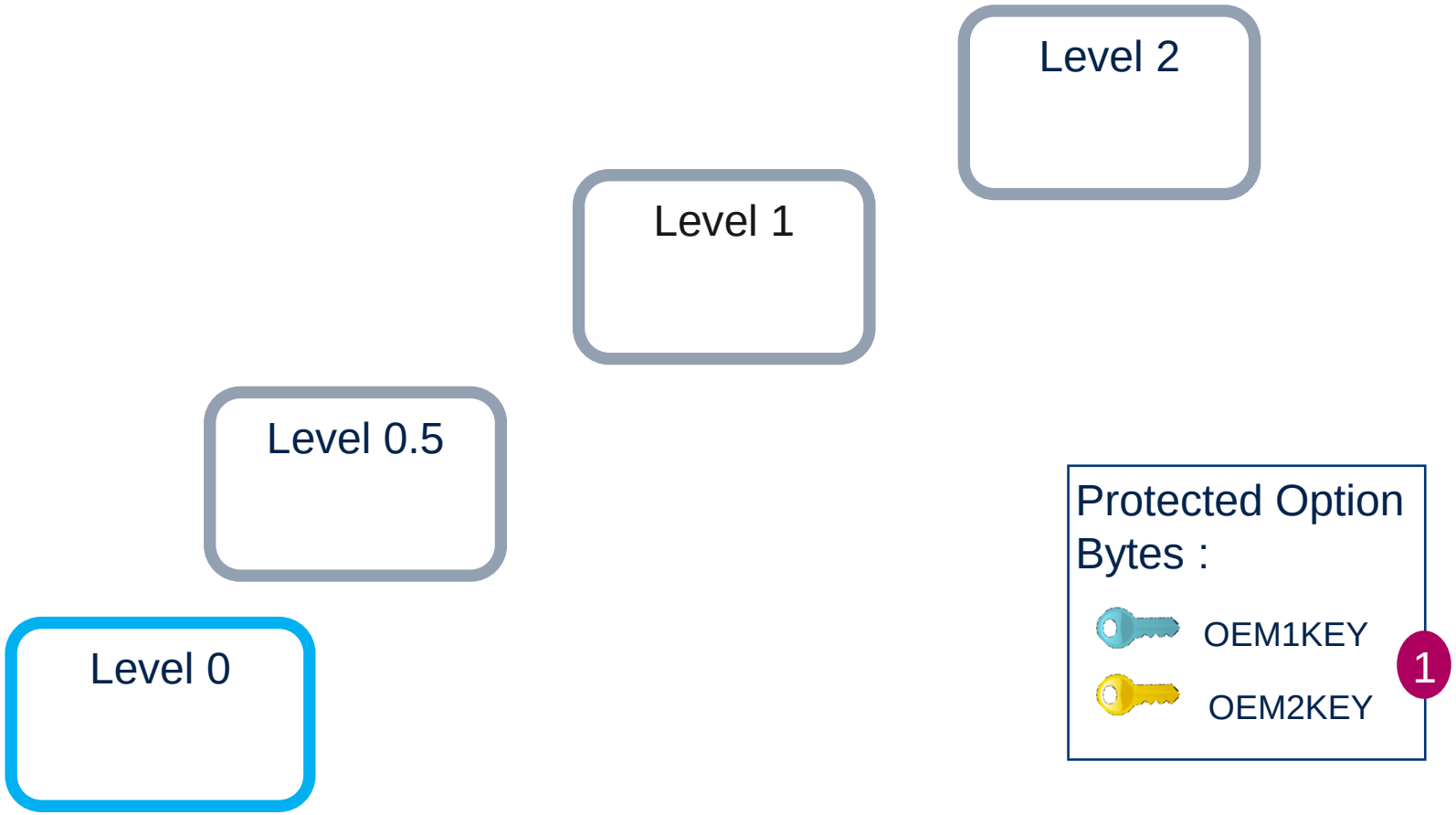
Step 6 : RDP level 0

Step 7 : Remove keys

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U5 Keyed RDP

STEP1



Step 1 : Provisioning

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OEMxLOCK status

7.9.7 FLASH non-secure status register (FLASH_NSSR)

This register is non-secure. It can be read and written by both secure and non-secure access. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

Address offset: 0x20

Reset value: 0x000X 0000

Access: no wait state; word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PD2	PD1	OEM2LOCK	OEM1LOCK	WDW	BSY
										r	r	r	r	r	r

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	OPTWERR	Res.	Res.	Res.	Res.	Res.	PGSERR	SIZERR	PGAERR	WRPERR	PROGERR	Res.	OPERR	EOP
		rc_w1						rc_w1	rc_w1	rc_w1	rc_w1	rc_w1		rc_w1	rc_w1

Bit 19 **OEM2LOCK**: OEM2 lock

This bit indicates that the OEM2 RDP key read during the OBL is not virgin. When set, the OEM2 RDP lock mechanism is active.

Bit 18 **OEM1LOCK**: OEM1 lock

This bit indicates that the OEM1 RDP key read during the OBL is not virgin. When set, the OEM1 RDP lock mechanism is active.

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Step 1 :Provisioning

Provision OEM1KEY key

- Check no key already provisioned: read FLASH_NSSR register
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-r32 0x40022020 4**
  - Result : 0x40022020 : 00000000
- Provision OEM1KEY key
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-lockRDP1 0x01234567 0x89ABCDEF**
  - Result: “Lock RDP1 password successfully done”
- Check key was provisioned
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-r32 0x40022020 4**
  - Result : 0x40022020 : 000**4**0000

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command_line.txt - Notepad

File Edit Format View Help

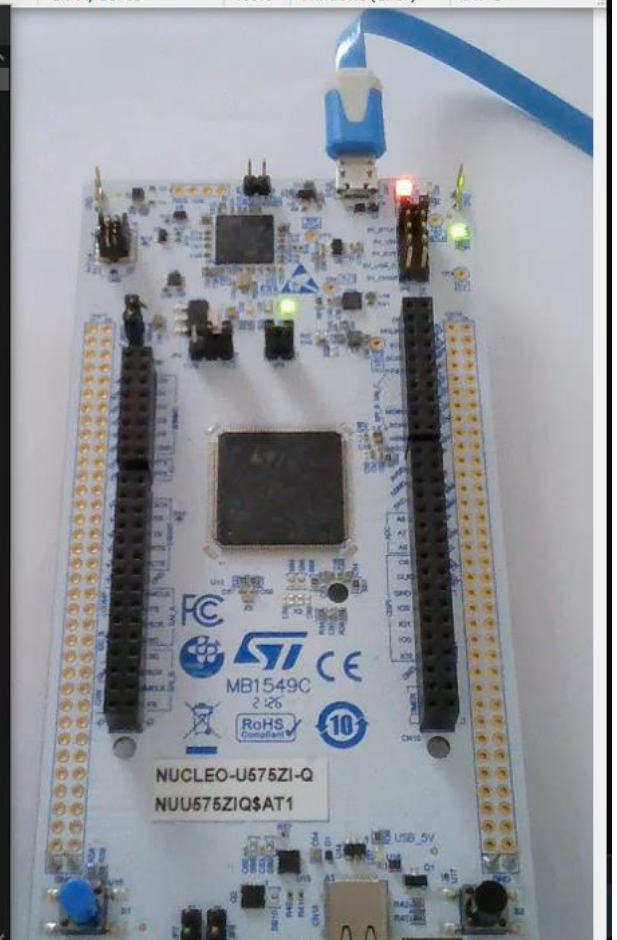
```
Read FLASH_NSSR register
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug -r32 0x40022020 4

Provision OEM1KEY key
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug  -lockRDP1 0x01234567 0x89ABCDEF
```

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Command Prompt

```
C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer\bin>
```



Step 1 :Provisioning

Provision OEM2KEY key

- Check no key already provisioned: Read flash register
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-r32 0x40022020 4**
  - Result : 0x40022020 : 000**4**0000
- Provision OEM2KEY key
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-lockRDP2 0xFEDCBA98 0x76543210**
  - Result: “Lock RDP2 password successfully done”
- Check key was provisioned
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-r32 0x40022020 4**
  - Result : 0x40022020 : 000**C**0000

*command_line.txt - Notepad

File Edit Format View Help

```
Read FLASH_NSSR register
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug -r32 0x40022020 4

Provision OEM1KEY key
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug  -lockRDP1 0x01234567 0x89ABCDEF

Provision OEM2KEY key
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug  -lockRDP2 0xFEDCBA98 0x76543210
```

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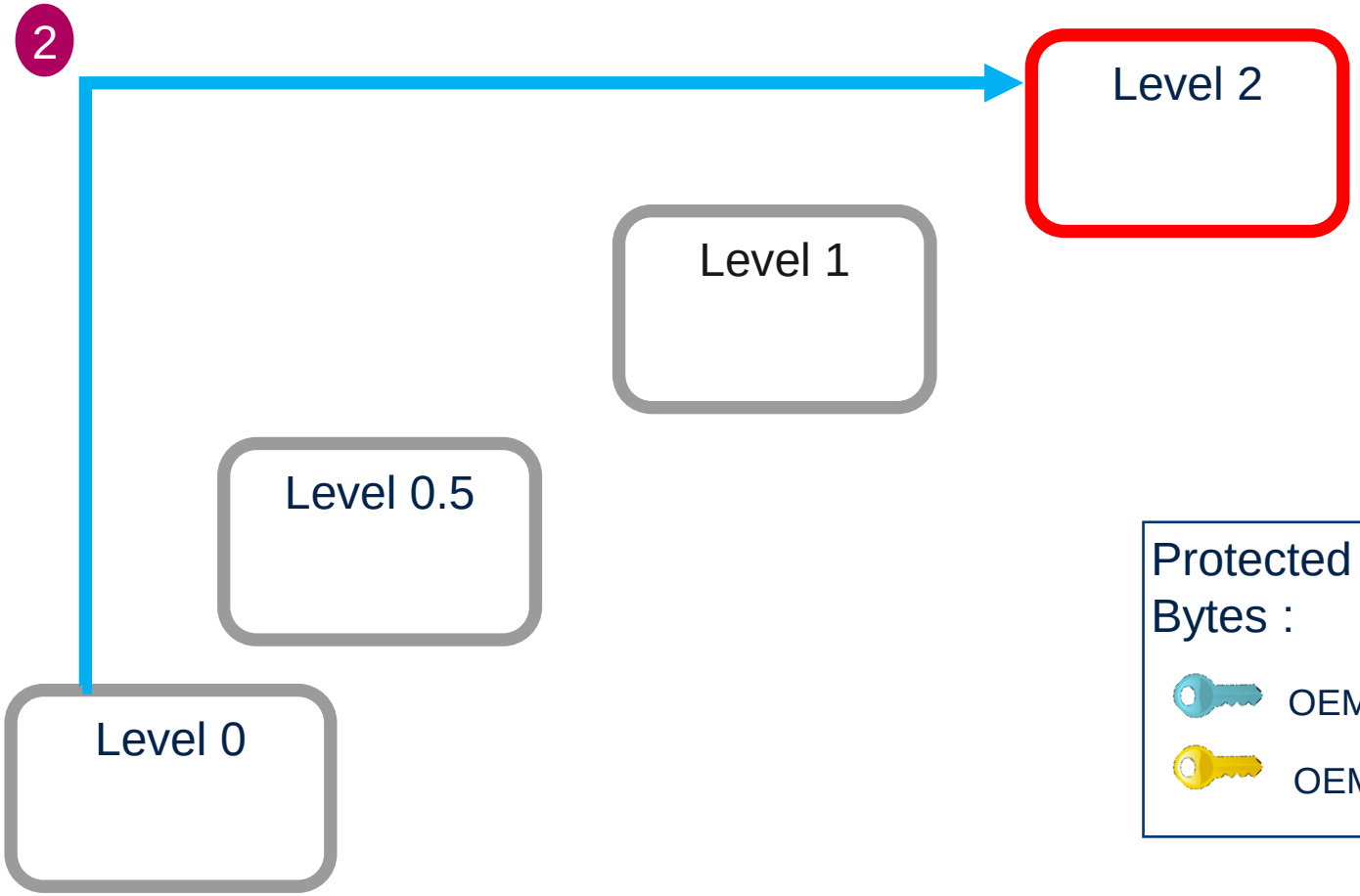
Command Prompt

```
C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer\bin>
```



U5 Keyed RDP

RDP transitions



Step 1 : Provisioning

Step 2 : RDP level 2

Protected Option Bytes :

OEM1KEY

OEM2KEY

Step 2 : RDP level 2

RDP 2 activation

- Enable RDP2
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-ob RDP=0xCC**
    Reconnection not possible at the end.
  - **POR => Led should blink**

- Get authentication ID
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **getauthid**
  - Result: DBGMCU_DBG_AUTH_DEVICE : 0xC6012CC2

command_line.txt - Notepad

File Edit Format View Help

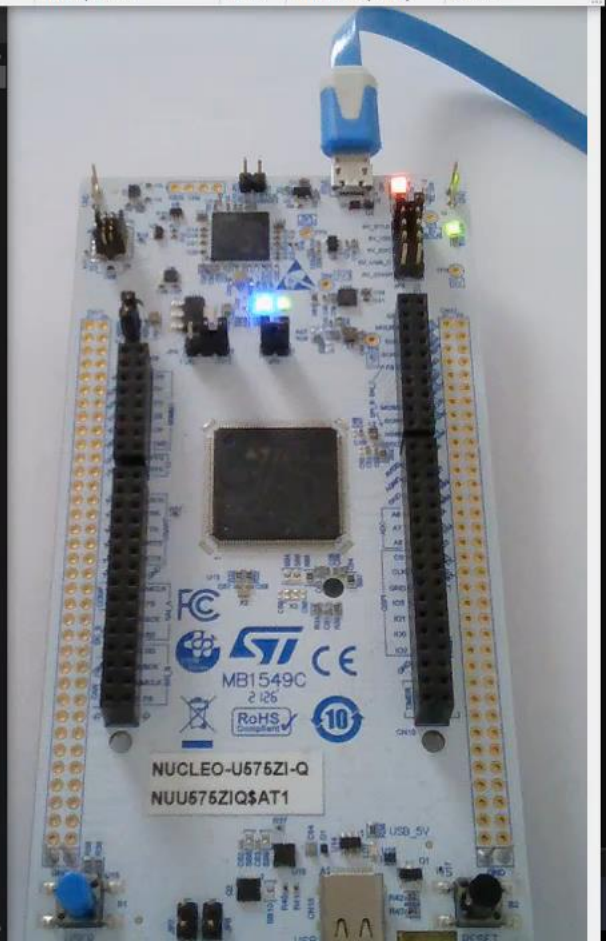
```
RDP level 2
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug  -ob RDP=0xCC

Get authentication ID
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug  getauthid
```

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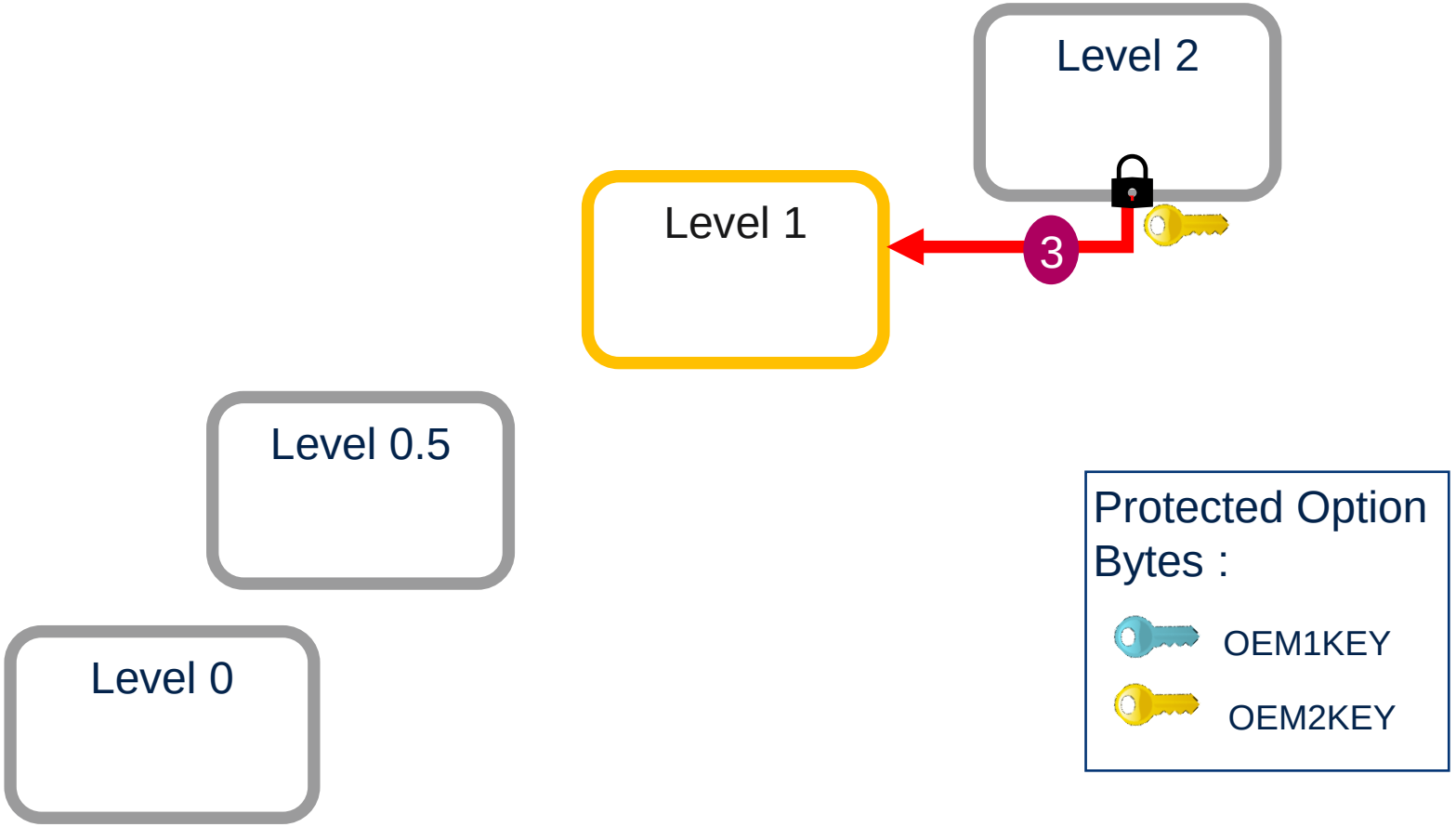
Command Prompt

```
C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer\bin>
```



Step 3 : RDP level 1

RDP transitions



Step 1 : Provisioning

Step 2 : RDP level 2

Step 3 : RDP level 1

Protected Option Bytes :

OEM1KEY

OEM2KEY

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Step 3 : RDP level 1

Regression to RDP1

- RDP2 to RDP1 regression:
  - STM32_Programmer_CLI.exe -c port=SWD **mode=UR -unlockRDP2 0xFEDCBA98 0x76543210**
  - Result : Unlock RDP2 password succefully done!
- Check RDP level :
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-ob displ**
  - **Result : RDP should 0xFF : the regression was done automatically**
- Check firmware status:
  - **POR: Led should blink: No mass erase when RDP2 to RDP1 regression**

command_line.txt - Notepad

File Edit Format View Help

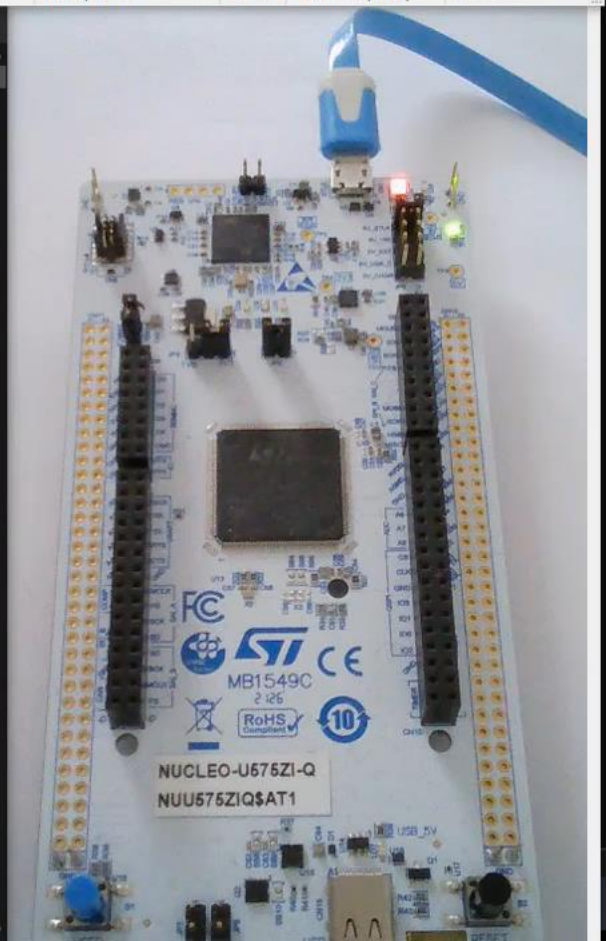
```
RDP2 to RDP1 regression
STM32_Programmer_CLI.exe -c port=SWD mode=UR -unlockRDP2 0xFEDCBA98 0x76543210

Check RDP level
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug -ob displ
```

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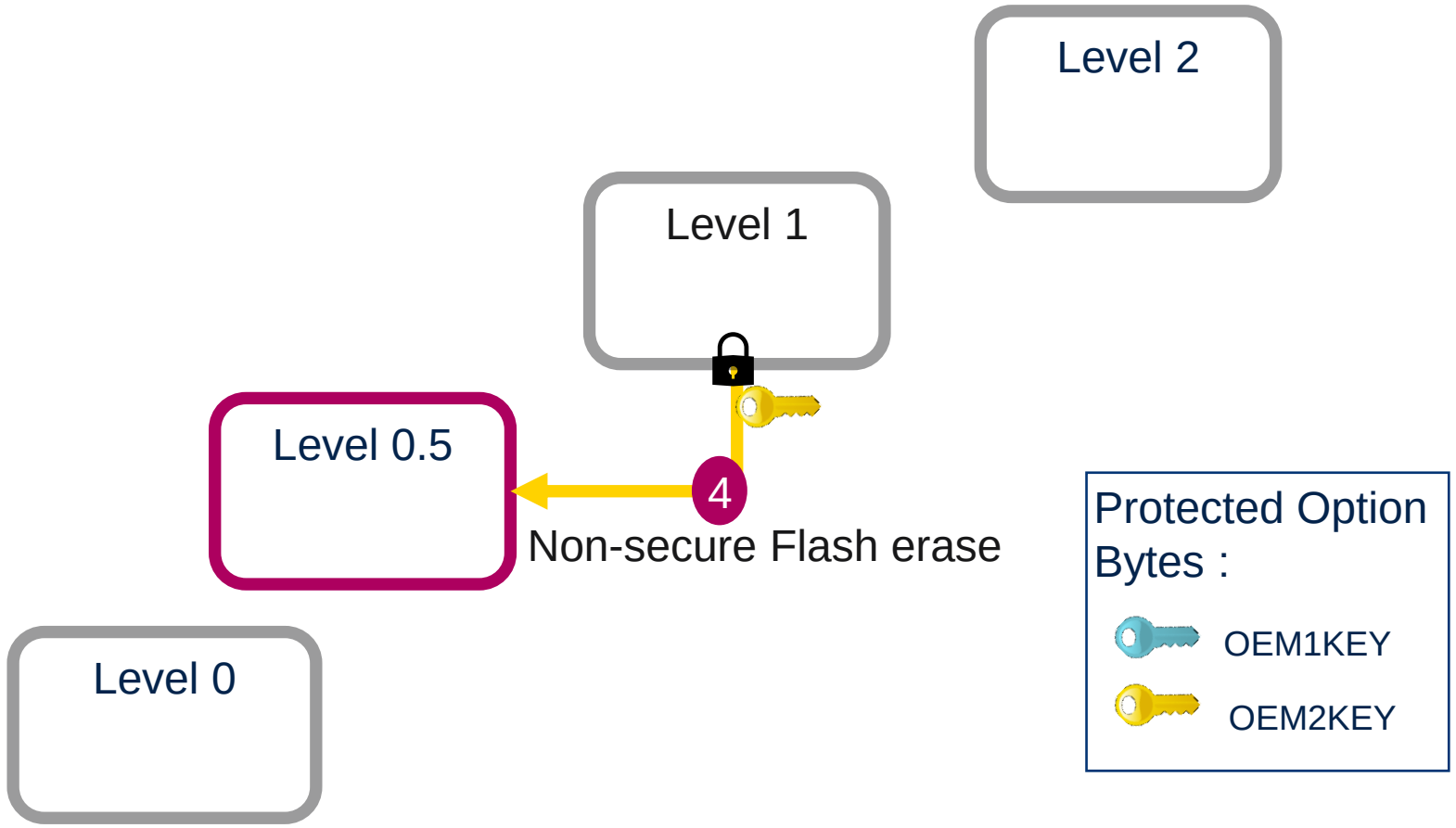
Command Prompt

```
C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer\bin>
```



U5 Keyed RDP

RDP transitions



Step 1 : Provisioning

Step 2 : RDP level 2

Step 3 : RDP level 1

Step 4 : RDP level 0.5
reflash non-secure App

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Step 4 : RDP level 0.5

Regression to RDP 0.5

- Unlock the RDP 0.5 regression:
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-unlockRDP1 0xFEDCBA98 0x76543210**
  - Expected result **:Unlock RDP1 password successfully done**
- Regression to RDP 0.5
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **–ob RDP=0x55**
  - Expected result **: POR: no Led blinking**
- Flash again the non secure
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **–d** \Training\MOOC_U5\MOOC1\Handson\GPIO_IOToggle_TrustZone_NonSecure.elf
  - Expected result **: POR: Led blinking**

command_line.txt - Notepad

File Edit Format View Help

```
Unlock the RDP 0.5 regression:
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug  -unlockRDP1 0xFEDCBA98 0x76543210

Regression to RDP 0.5
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug  -ob RDP=0x55

Flash again the non secure
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug  -d \Training\MOOC_U5\MOOC1\Handson\GPIO_IOToggle_TrustZone_NonSecure.elf
```

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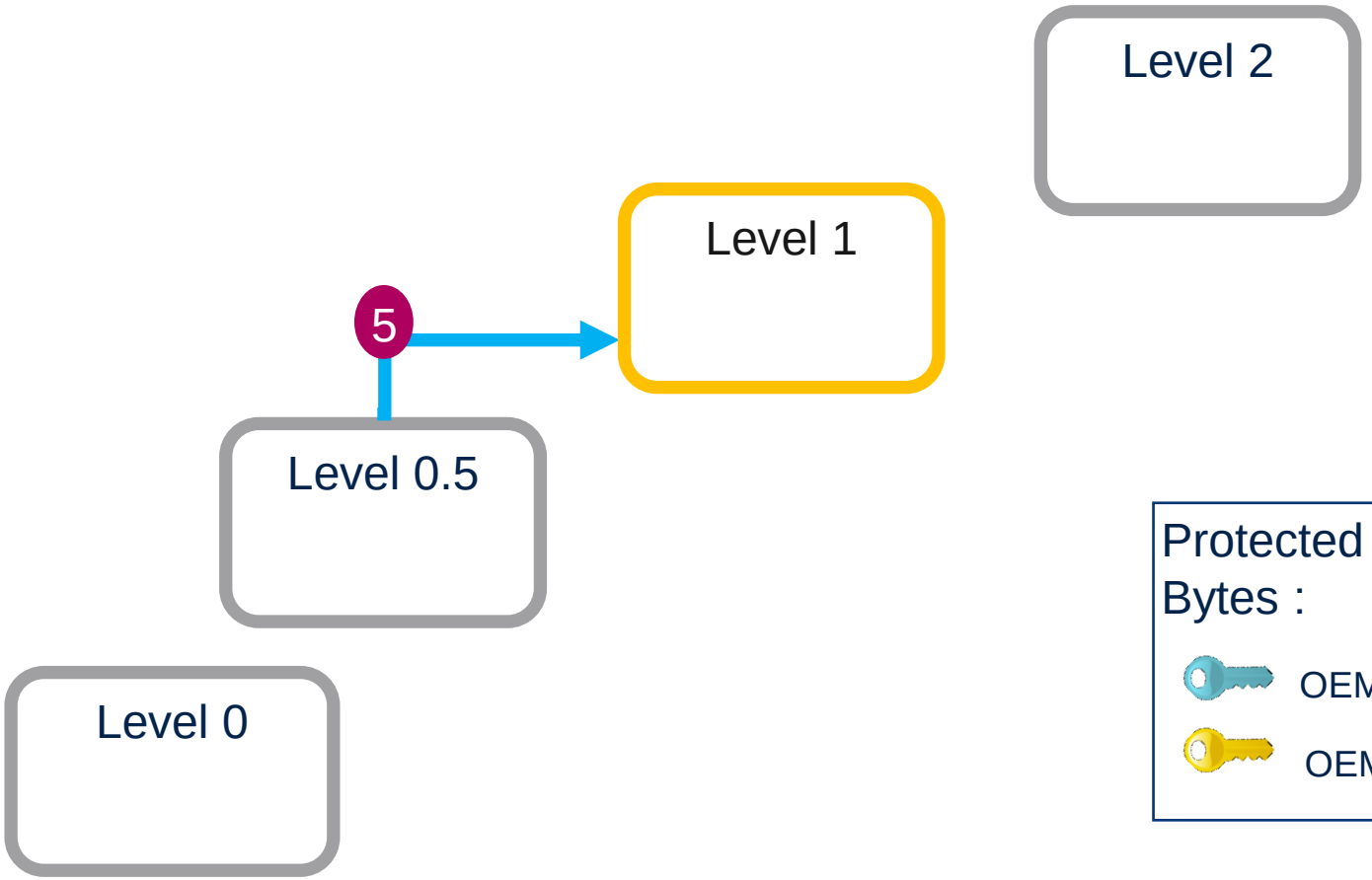
Command Prompt

```
C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer\bin>
```



U5 Keyed RDP

RDP transitions



Protected Option Bytes :

- OEM1KEY
- OEM2KEY

Step 1 : Provisioning

Step 2 : RDP level 2

Step 3 : RDP level 1

Step 4 : RDP level 0.5
reflash non-secure App

Step 5 : RDP level 1

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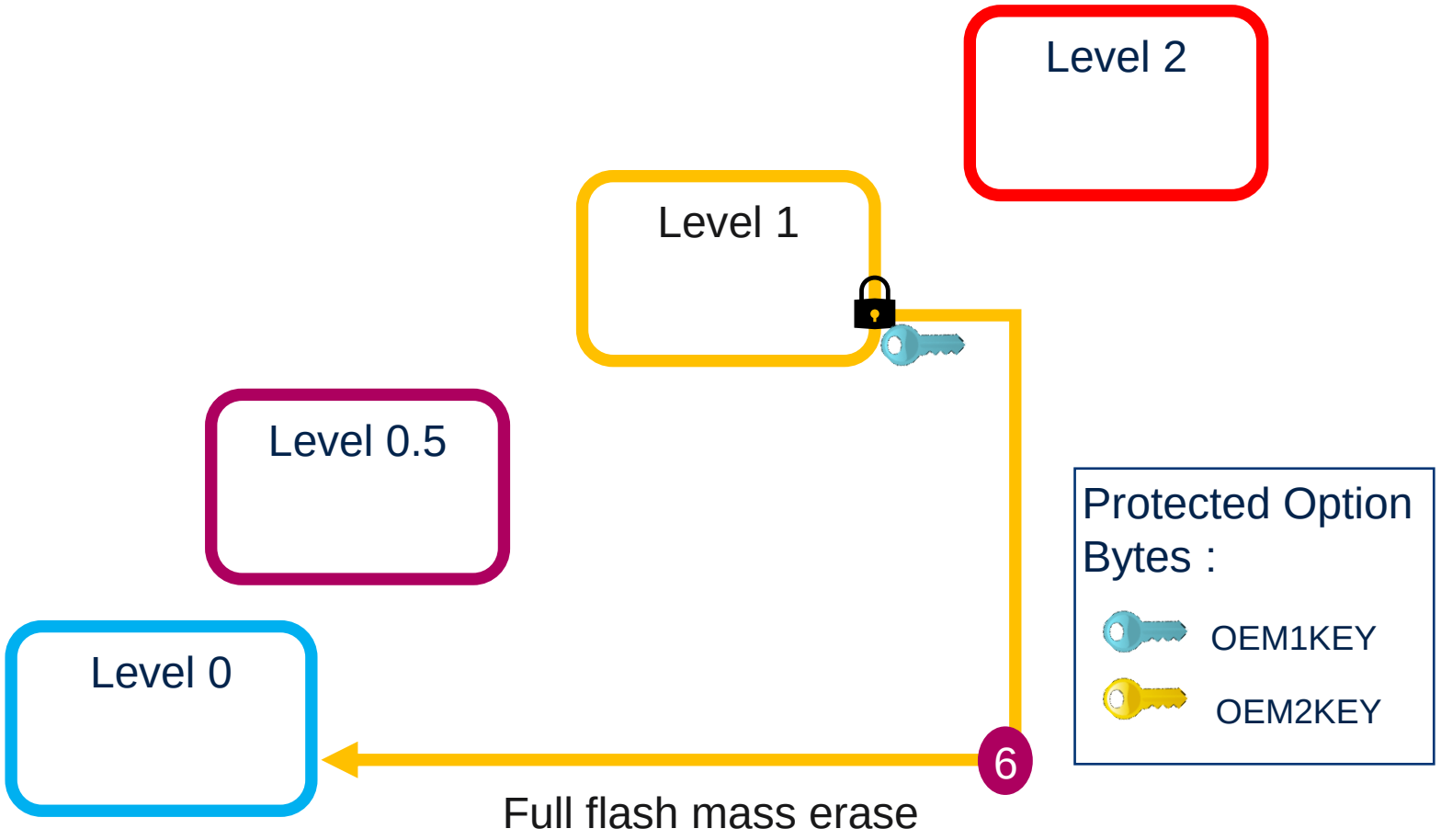
Step 5 : RDP level 1

RDP 0.5 - RDP1

- RDP 0.5 to RDP1
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-ob RDP=0xDC**
  - Result : Reconnection not possible at the end. **POR: Led should blink**

U5 Keyed RDP

RDP transitions



Step 1 : Provisioning

Step 2 : RDP level 2

Step 3 : RDP level 1

Step 4 : RDP level 0.5 reflash non-secure App

Step 5 : RDP level 1

Step 6 : RDP level 0

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Step 6 : RDP level 0

Regression to RDP0

- Unlock the RDP 0 regression :
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-unlockRDP1 0x01234567 0x89ABCDEF**
  - Expected result **:Unlock RDP1 password successfully done**
- RDP 0 regression
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-ob RDP=0xAA**
  - Expected result **:** RDP back to 0xAA + mass erase done

*command_line.txt - Notepad

File Edit Format View Help

```
RDP 0.5 to RDP1
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug  -ob RDP=0xDC

Unlock the RDP 0 regression
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug   -unlockRDP1 0x01234567 0x89ABCDEF

RDP 0 regression
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug  -ob RDP=0xAA
```

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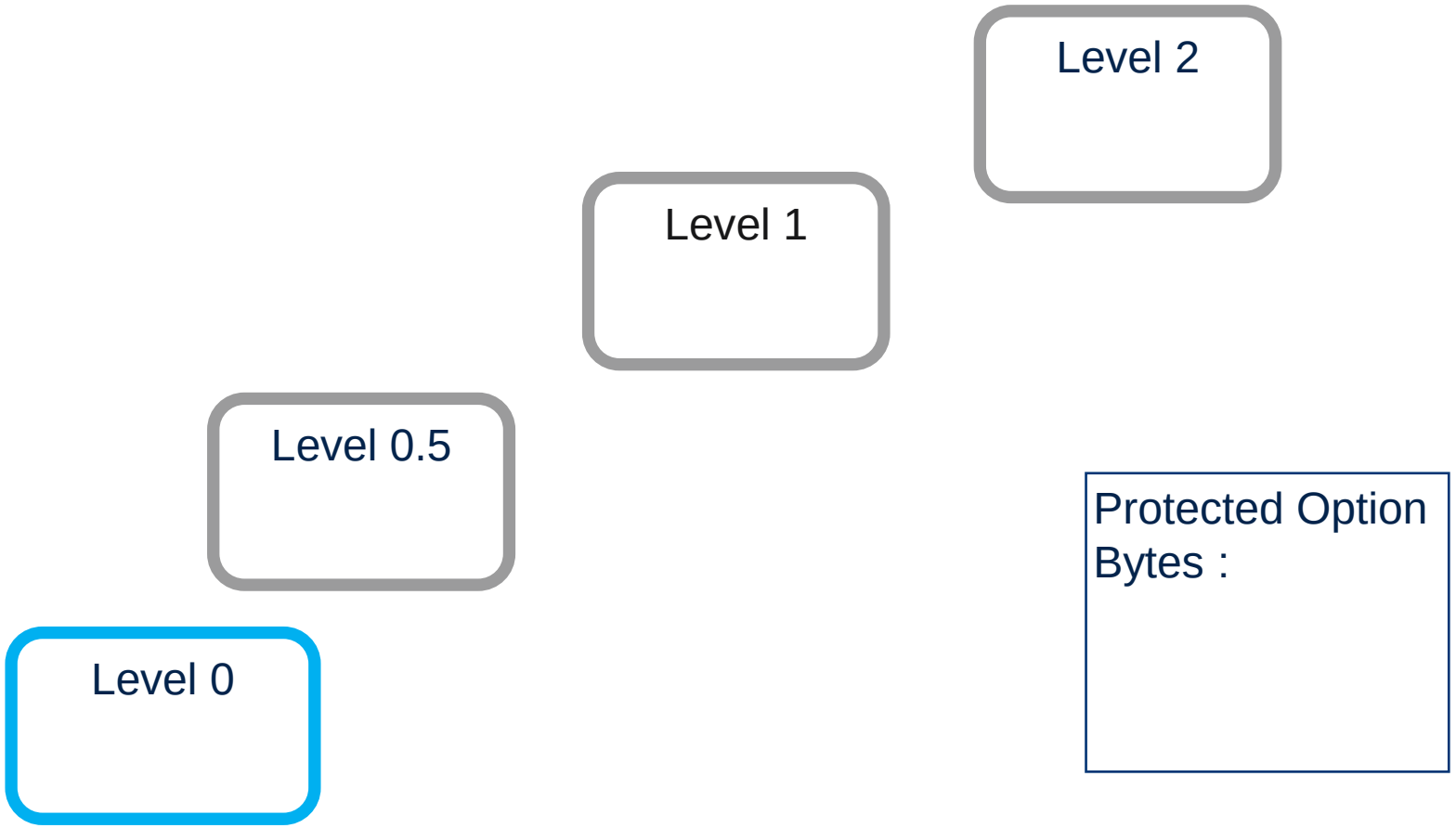
Command Prompt

```
C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer\bin>
```



U5 Keyed RDP

RDP transitions



Step 1 : Provisioning

Step 2 : RDP level 2

Step 3 : RDP level 1

Step 4 : RDP level 0.5
reflash non-secure App

Step 5 : RDP level 1

Step 6 : RDP level 0

Step 7 : Remove keys

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Step 6 : Remove keys

Remove OEM1KEY

- One important feature is to be able to remove a key
- Erase OEM1KEY
  - Check status
    STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-r32 0x40022020 4**
    Expected result : 0x40022020 : 000**C**0000
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-lockRDP1 0xFFFFFFFF 0xFFFFFFFF**
  - Check status
    STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-r32 0x40022020 4**
    Expected result : 0x40022020 : 000**8**0000 (Only OEM2key is remaining)

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Step 6 : Remove keys

Remove OEM2KEY

- Erase OEM2KEY
  - STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-lockRDP2 0xFFFFFFFF 0xFFFFFFFF**
  - Check status
    STM32_Programmer_CLI.exe -c port=SWD mode=hotplug **-r32 0x40022020 4**
    Expected result : 0x40022020 : 00000000 (back to legacy RDP)

*command_line.txt - Notepad

File Edit Format View Help

```
Read FLASH_NSSR register
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug -r32 0x40022020 4

Erase OEM1KEY
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug  -lockRDP1 0xFFFFFFFF 0xFFFFFFFF

Erase OEM2KEY
STM32_Programmer_CLI.exe -c port=SWD mode=hotplug  -lockRDP2 0xFFFFFFFF 0xFFFFFFFF
```

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Command Prompt

```
C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer\bin>
```



Useful documentation

U5 keyed RDP

- RM0456 : Reference manual STM32U575/585 Arm®-based 32-bit MCUs
- AN5347 : Arm® TrustZone® features for STM32L5 and STM32U5 Series

Conclusion

U5 keyed RDP

- U5 Keyed RDP allows to control the regression of the RDP from any level (including RDP level 2) and in a secure way
- This new feature doesn’t decrease the flash content protection.
- We have seen how easy this could be done thanks STM32CubeProgrammer tool.

Our technology starts with You



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